

Material Safety Data Sheet

Petroleum Ether

ACC# 18330

Section 1 - Chemical Product and Company Identification

MSDS Name: Petroleum Ether

Catalog Numbers: E120-4, E120-400, E1204LOT001, E1204LOT002, E1204LOT003, E120550, E12055115, E12055200, E120552B, E1205550, E120J4, E120POP19, E120POP200, E120POP50, E120SS19, E139-1, E139-20, E139-200, E139-4, E139-500, E1394LC, E139FB115, E139FB19, E139FB200, E139FB28, E139FB50, E139J4, E139POP19, E139POP200, E139POP50, E139POPB19, E139POPB200, E139POPB50, E139RB115, E139RB19, E139RB200, E139RB50, E139RS115, E139RS200, E139RS28, E139RS50, E139S-4, E139SK-4, E139SS115, E139SS19, E139SS200, E139SS28, E139SS50, FLP480RS-115, FLP480RS-200, FLP480RS-28, FLP480RS-50, FLP481RS-200, NC9480472, P480-4, P4804LC, P4804LOT001, P480POP19, P480POP200, P480POP50, P480RS115, P480RS200, P480RS28, P480RS50, P480SS115, P480SS19, P480SS200, P480SS28, P480SS50, P481POP200, P481SS-200

Synonyms: Naphtha Solvent; Naphtha Petroleum

Company Identification:

Fisher Scientific
1 Reagent Lane
Fair Lawn, NJ 07410

For information, call: 201-796-7100

Emergency Number: 201-796-7100

For CHEMTREC assistance, call: 800-424-9300

For International CHEMTREC assistance, call: 703-527-3887

Section 2 - Composition, Information on Ingredients

CAS#	Chemical Name	Percent	EINECS/ELINCS
68476-50-6	Hydrocarbons, Includes chemicals below	~100	270-690-8
109-66-0	Pentane	<73	203-692-4
107-83-5	Isohexane	<11.0	203-523-4
96-14-0	3-methylpentane	<6.0	202-481-4
75-83-2	Neohexane	<4.0	200-906-8
79-29-8	2,3-dimethylbutane	<4.0	201-193-6
78-78-4	Iso-pentane	<2.0	201-142-8
287-92-3	Cyclopentane	<2.0	206-016-6
Not available	Various Isomers	<1.0	unlisted

Hazard Symbols: T F N

Risk Phrases: 11 36/38 45 51/53 65 67

Section 3 - Hazards Identification

EMERGENCY OVERVIEW

Appearance: colourless liquid. Flash Point: -50 deg F. **Danger!** Aspiration hazard if swallowed. Can enter lungs and cause damage. May cause central nervous system depression. Causes digestive and respiratory tract irritation. Causes eye and skin irritation. Poison! Vapor harmful. Harmful or fatal if swallowed. **Flammable liquid and vapor.** Cancer hazard.

Target Organs: Kidneys, central nervous system, lungs, eyes, skin.

Potential Health Effects

Eye: Causes eye irritation. May cause conjunctivitis and corneal inflammation.

Skin: Exposure may cause irritation characterized by redness, dryness, and inflammation. Prolonged exposure may produce blisters. May aggravate existing skin disorders.

Ingestion: Aspiration hazard. Causes gastrointestinal irritation with nausea, vomiting and diarrhea. May cause central nervous system depression, characterized by excitement, followed by headache, dizziness, drowsiness, and nausea. Advanced stages may cause collapse, unconsciousness, coma and possible death due to respiratory failure.

Inhalation: Inhalation of high concentrations may cause central nervous system effects characterized by nausea, headache, dizziness, unconsciousness and coma. High vapor concentrations may cause drowsiness. Aspiration may cause respiratory swelling and pneumonitis.

Chronic: Prolonged or repeated skin contact may cause dermatitis. May cause kidney damage. Cancer suspect agent.

Section 4 - First Aid Measures

Eyes: Immediately flush eyes with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Get medical aid.

Skin: Get medical aid. Immediately flush skin with plenty of soap and water for at least 15 minutes while removing contaminated clothing and shoes. Wash clothing before reuse.

Ingestion: Do NOT induce vomiting. If victim is conscious and alert, give 2-4 cupfuls of milk or water. Never give anything by mouth to an unconscious person. Get medical aid immediately.

Inhalation: Get medical aid immediately. Remove from exposure to fresh air immediately. If breathing is difficult, give oxygen. Do NOT use mouth-to-mouth resuscitation. If breathing has ceased apply artificial respiration using oxygen and a suitable mechanical device such as a bag and a mask.

Notes to Physician: Gastric lavage by qualified medical personnel may be considered depending on quantity of material ingested.

Section 5 - Fire Fighting Measures

General Information: As in any fire, wear a self-contained breathing apparatus in pressure-demand, MSHA/NIOSH (approved or equivalent), and full protective gear. Vapors may form an explosive mixture with air. Vapors can travel to a source of ignition and flash back. Use water spray to keep fire-exposed containers cool. Vapor may cause flash fire. Vapors may be heavier than air. They can spread along the ground and collect in low or confined areas. Containers may explode when heated.

Extinguishing Media: Use water spray to cool fire-exposed containers. Water may be ineffective. This material is lighter than water and insoluble in water. The fire could easily be spread by the use of water in an area where the water cannot be contained. Do NOT use straight streams of water. For small fires, use dry chemical, carbon dioxide, water spray or regular foam. Cool containers with flooding quantities of water until well after fire is out. For large fires, use water spray, fog or regular foam.

Flash Point: -50e deg F (-45.56 deg C)

Autoignition Temperature: 500 deg F (260.00 deg C)

Explosion Limits, Lower:1.1

Upper: 5.9

NFPA Rating: (estimated) Health: 2; Flammability: 4; Instability: 0

Section 6 - Accidental Release Measures

General Information: Use proper personal protective equipment as indicated in Section 8.

Spills/Leaks: Absorb spill with inert material (e.g. vermiculite, sand or earth), then place in suitable container. Avoid runoff into storm sewers and ditches which lead to waterways. Clean up spills immediately, observing precautions in the Protective Equipment section. Remove all sources of ignition. Use a spark-proof tool. A vapor suppressing foam may be used to reduce vapors. Water spray may reduce vapor but may not prevent ignition in closed spaces.

Section 7 - Handling and Storage

Handling: Wash thoroughly after handling. Use with adequate ventilation. Ground and bond containers when transferring material. Avoid contact with eyes, skin, and clothing. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep container tightly closed. Avoid contact with heat, sparks and flame. Avoid ingestion and inhalation. Prevent build up of vapors to explosive concentration. Do not pressurize, cut, weld, braze, solder, drill, grind, or expose empty

containers to heat, sparks or open flames.

Storage: Keep away from heat, sparks, and flame. Keep away from sources of ignition. Store in a tightly closed container. Store in a cool, dry, well-ventilated area away from incompatible substances. Flammables-area.

Section 8 - Exposure Controls, Personal Protection

Engineering Controls: Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower. Use adequate general or local exhaust ventilation to keep airborne concentrations below the permissible exposure limits.

Exposure Limits

Chemical Name	ACGIH	NIOSH	OSHA - Final PELs
Hydrocarbons, Includes chemicals below	none listed	none listed	none listed
Pentane	600 ppm TWA	120 ppm TWA; 350 mg/m ³ TWA 1500 ppm IDLH (10 percent lower explosive limit)	1000 ppm TWA; 2950 mg/m ³ TWA
Isohexane	none listed	none listed	none listed
3-methylpentane	none listed	none listed	none listed
Neohexane	none listed	none listed	none listed
2,3-dimethylbutane	none listed	none listed	none listed
Iso-pentane	600 ppm TWA (Pentane, all isomers)	none listed	none listed
Cyclopentane	600 ppm TWA	600 ppm TWA; 1720 mg/m ³ TWA	none listed
Various Isomers	none listed	none listed	none listed

OSHA Vacated PELs: Hydrocarbons, Includes chemicals below: No OSHA Vacated PELs are listed for this chemical. Pentane: 600 ppm TWA; 1800 mg/m³ TWA; 750 ppm STEL; 2250 mg/m³ STEL Isohexane: No OSHA Vacated PELs are listed for this chemical. 3-methylpentane: No OSHA Vacated PELs are listed for this chemical. Neohexane: No OSHA Vacated PELs are listed for this chemical. 2,3-dimethylbutane: No OSHA Vacated PELs are listed for this chemical. Iso-pentane: No OSHA Vacated PELs are listed for this chemical. Cyclopentane: 600 ppm TWA; 1720 mg/m³ TWA Various Isomers: No OSHA Vacated PELs are listed for this chemical.

Personal Protective Equipment

Eyes: Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

Skin: Wear appropriate protective gloves to prevent skin exposure.

Clothing: Wear appropriate protective clothing to prevent skin exposure.

Respirators: Follow the OSHA respirator regulations found in 29CFR 1910.134 or European Standard EN 149. Always use a NIOSH or European Standard EN 149 approved respirator when necessary.

Section 9 - Physical and Chemical Properties

Physical State: Liquid

Appearance: colourless

Odor: mild odor - gasoline-like

pH: Not available.

Vapor Pressure: 485 mm Hg @ 20C

Vapor Density: 2.5

Evaporation Rate: >1.0 (butyl acetate=1)

Viscosity: Not available.

Boiling Point: 37-56C

Freezing / Melting Point: Not available.

Decomposition Temperature: Not available.

Solubility: Insoluble.

Specific Gravity / Density: 0.74

Molecular Formula: Hydrocarbon

Molecular Weight: Not available

Section 10 - Stability and Reactivity

Chemical Stability: Stable under normal temperatures and pressures.

Conditions to Avoid: Incompatible materials, ignition sources, excess heat.

Incompatibilities with Other Materials: May explode with nitrogen tetroxide, potential violent reaction with strong oxidizers.

Hazardous Decomposition Products: Carbon monoxide, carbon dioxide.

Hazardous Polymerization: Has not been reported.

Section 11 - Toxicological Information

RTECS#:

CAS# 68476-50-6 unlisted.

CAS# 109-66-0: RZ9450000

CAS# 107-83-5: SA2985000

CAS# 96-14-0 unlisted.

CAS# 75-83-2 unlisted.

CAS# 79-29-8: EJ9350000

CAS# 78-78-4: EK4430000

CAS# 287-92-3: GY2390000

CAS# Not available unlisted.

LD50/LC50:

Not available.

CAS# 109-66-0:

Inhalation, rat: LC50 = 364 gm/m³/4H;

Oral, rat: LD50 = 400 mg/kg;

CAS# 107-83-5:

CAS# 96-14-0:

CAS# 75-83-2:

CAS# 79-29-8:

CAS# 78-78-4:

CAS# 287-92-3:

Inhalation, mouse: LC50 = 72 gm/m³;

Inhalation, rat: LC50 = 106 gm/m³;

Oral, mouse: LD50 = 12800 mg/kg;

Oral, rat: LD50 = 11400 mg/kg;

Carcinogenicity:

CAS# 68476-50-6: Not listed by ACGIH, IARC, NIOSH, NTP, or OSHA. CAS# 109-66-0: Not listed by ACGIH, IARC, NIOSH, NTP, or OSHA. CAS# 107-83-5: Not listed by ACGIH, IARC, NIOSH, NTP, or OSHA. CAS# 96-14-0: Not listed by ACGIH, IARC, NIOSH, NTP, or OSHA. CAS# 75-83-2: Not listed by ACGIH, IARC, NIOSH, NTP, or OSHA. CAS# 79-29-8: Not listed by ACGIH, IARC, NIOSH, NTP, or OSHA. CAS# 78-78-4: Not listed by ACGIH, IARC, NIOSH, NTP, or OSHA. CAS# 287-92-3: Not listed by ACGIH, IARC, NIOSH, NTP, or OSHA. CAS# Not available: Not listed by ACGIH, IARC, NIOSH, NTP, or OSHA.

Epidemiology: Epidemiological studies involving petroleum refinery workers indicate persons with routine exposure to petroleum or one of its constituents may be at an increased risk to the development of benign neoplasms, digestive tract cancer, and skin cancer.

Teratogenicity: No information available.

Reproductive Effects: No information available.

Neurotoxicity: No information available.

Mutagenicity: No information available.

Other Studies: See actual entry in RTECS for complete information.

Section 12 - Ecological Information

Ecotoxicity: No data available. This chemical is expected to cause some oxygen depletion in aquatic systems. It has a low potential to affect aquatic systems. It has a low potential to affect aquatic organisms, secondary waste treatment microorganisms and the germination of some plants. It has a moderate potential to affect the germination and growth of

some plants.

Environmental: This chemical does not bioconcentrate. When diluted with water, this chemical released directly or indirectly into the environment is not expected to have a significant impact.

Physical: No information found.

Other: May be toxic to aquatic organisms; May cause long-term adverse effects in the aquatic environment.

Section 13 - Disposal Considerations

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. US EPA guidelines for the classification determination are listed in 40 CFR Parts 261.3. Additionally, waste generators must consult state and local hazardous waste regulations to ensure complete and accurate classification.

RCRA P-Series: None listed.

RCRA U-Series: None listed.

Section 14 - Transport Information

	US DOT	IATA	RID/ADR	IMO	Canada TDG
Shipping Name:	PETROLEUM DISTILLATES, N.O.S.				No information available.
Hazard Class:	3				
UN Number:	UN1268				
Packing Group:	II				

Section 15 - Regulatory Information

US FEDERAL

TSCA

CAS# 68476-50-6 is listed on the TSCA inventory.

CAS# 109-66-0 is listed on the TSCA inventory.

CAS# 107-83-5 is listed on the TSCA inventory.

CAS# 96-14-0 is listed on the TSCA inventory.

CAS# 75-83-2 is listed on the TSCA inventory.

CAS# 79-29-8 is listed on the TSCA inventory.

CAS# 78-78-4 is listed on the TSCA inventory.

CAS# 287-92-3 is listed on the TSCA inventory.

Various Isomers is not listed on the TSCA inventory. It is for research and development use only.

Health & Safety Reporting List

None of the chemicals are on the Health & Safety Reporting List.

Chemical Test Rules

None of the chemicals in this product are under a Chemical Test Rule.

Section 12b

None of the chemicals are listed under TSCA Section 12b.

TSCA Significant New Use Rule

None of the chemicals in this material have a SNUR under TSCA.

SARA

Section 302 (RQ)

None of the chemicals in this material have an RQ.

Section 302 (TPQ)

None of the chemicals in this product have a TPQ.

SARA Codes

CAS # 109-66-0: acute, chronic, flammable. CAS # 107-83-5: acute, flammable. CAS # 96-14-0: acute, flammable. CAS # 78-78-4: acute, flammable. CAS # 287-92-3: flammable.

Section 313

No chemicals are reportable under Section 313.

Clean Air Act:

This material does not contain any hazardous air pollutants. This material does not contain any Class 1 Ozone depleters. This material does not contain any Class 2 Ozone depleters.

Clean Water Act:

None of the chemicals in this product are listed as Hazardous Substances under the CWA. None of the chemicals in this product are listed as Priority Pollutants under the CWA. None of the chemicals in this product are listed as Toxic Pollutants under the CWA.

OSHA:

None of the chemicals in this product are considered highly hazardous by OSHA.

STATE

CAS# 68476-50-6 is not present on state lists from CA, PA, MN, MA, FL, or NJ.

CAS# 109-66-0 can be found on the following state right to know lists: California, New Jersey, Florida, Pennsylvania, Minnesota, Massachusetts.

CAS# 107-83-5 can be found on the following state right to know lists: Florida, Pennsylvania, Minnesota, Massachusetts.

CAS# 96-14-0 can be found on the following state right to know lists: Florida, Pennsylvania, Massachusetts.

CAS# 75-83-2 can be found on the following state right to know lists: New Jersey, Florida, Pennsylvania, Massachusetts.

CAS# 79-29-8 can be found on the following state right to know lists: New Jersey, Florida, Pennsylvania, Massachusetts.

CAS# 78-78-4 can be found on the following state right to know lists: New Jersey, Florida, Pennsylvania, Massachusetts.

CAS# 287-92-3 can be found on the following state right to know lists: California, New Jersey, Florida, Pennsylvania, Minnesota, Massachusetts.

CAS# Not available is not present on state lists from CA, PA, MN, MA, FL, or NJ.

California No Significant Risk Level: None of the chemicals in this product are listed.

European / International Regulations**European Labeling in Accordance with EC Directives****Hazard Symbols:**

T F N

Risk Phrases:

R 11 Highly flammable.

R 36/38 Irritating to eyes and skin.

R 45 May cause cancer.

R 51/53 Toxic to aquatic organisms; may cause long-term adverse effects in the aquatic environment.

R 65 Harmful: may cause lung damage if swallowed.

R 67 Vapors may cause drowsiness and dizziness.

Safety Phrases:

S 16 Keep away from sources of ignition - No smoking.

S 29 Do not empty into drains.

S 33 Take precautionary measures against static discharges.

S 45 In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible).

S 53 Avoid exposure - obtain special instructions before use.

S 9 Keep container in a well-ventilated place.

S 61 Avoid release to the environment. Refer to special instructions/Safety data sheets.

S 62 If swallowed, do not induce vomiting: seek medical advice immediately and show this container or label.

WGK (Water Danger / Protection)

CAS# 68476-50-6: No information available.

CAS# 109-66-0: 1

CAS# 107-83-5: 1

CAS# 96-14-0: 1

CAS# 75-83-2: No information available.

CAS# 79-29-8: No information available.

CAS# 78-78-4: 1

CAS# 287-92-3: 1

CAS# Not available: No information available.

Canada - DSL / NDSL

CAS# 109-66-0 is listed on Canada's DSL List.
CAS# 107-83-5 is listed on Canada's DSL List.
CAS# 96-14-0 is listed on Canada's DSL List.
CAS# 75-83-2 is listed on Canada's DSL List.
CAS# 79-29-8 is listed on Canada's DSL List.
CAS# 78-78-4 is listed on Canada's DSL List.
CAS# 287-92-3 is listed on Canada's DSL List.
CAS# 68476-50-6 is listed on Canada's NDSL List.

Canada - WHMIS

This product has a WHMIS classification of B2, D2B.

Canadian Ingredient Disclosure List

CAS# 109-66-0 is listed on the Canadian Ingredient Disclosure List.
CAS# 107-83-5 is listed on the Canadian Ingredient Disclosure List.
CAS# 75-83-2 is listed on the Canadian Ingredient Disclosure List.
CAS# 79-29-8 is listed on the Canadian Ingredient Disclosure List.
CAS# 287-92-3 is listed on the Canadian Ingredient Disclosure List.

Exposure Limits

CAS# 109-66-0: OEL-AUSTRALIA:TWA 600 ppm (1800 mg/m3);STEL 750 ppm (2250 mg/m3) OEL-BELGIUM:TWA 600 ppm (1770 mg/m3);STEL 750 ppm (2210 mg/m3) OEL-DENMARK:TWA 500 ppm (1500 mg/m3) OEL-FINLAND:TWA 500 ppm (1500 mg/m3);STEL 625 ppm (1800 mg/m3) OEL-FRANCE:TWA 600 ppm (1800 mg /m3) OEL-GERMANY:TWA 1000 ppm (2950 mg/m3) OEL-HUNGARY:TWA 500 mg/m3 ;STEL 1500 mg/m3 OEL-JAPAN:TWA 300 ppm (880 mg/m3) OEL-THE NETHERLANDS:TWA 600 ppm (1800 mg/m3) JAN9 OEL-THE PHILIPPINES:TWA 1000 ppm (2950 mg/m3) OEL-POLAND:TWA 100 mg/m3 OEL-RUSSIA:TWA 300 ppm;STEL 300 mg/m3 OEL-SWEDEN:TWA 600 ppm (1800 mg/m3);STEL 750 ppm (2000 mg/m3) OEL-SWITZERLAND:TWA 600 ppm (1800 mg/m3) OEL-TURKEY:TWA 1000 ppm (2950 mg/m3) OEL-UNITED KINGDOM:TWA 600 ppm (1800 mg/m3);STEL 750 ppm OEL IN BULGARIA, COLOMBIA, JORDAN, KOREA check ACGIH TLV OEL IN NEW ZEALAND, SINGAPORE, VIETNAM check ACGI TLV
CAS# 78-78-4: OEL-DENMARK:TWA 500 ppm (1500 mg/m3) OEL IN BULGARIA, COLOMBIA, JORDAN, KOREA check ACGIH TLV OEL IN NEW ZEALAND, SINGAPORE, VIETNAM check ACGI TLV
CAS# 287-92-3: OEL-AUSTRALIA:TWA 600 ppm (1720 mg/m3) OEL-BELGIUM:TWA 600 ppm (1720 mg/m3) OEL-DENMARK:TWA 300 ppm (850 mg/m3) OEL-FRANCE:TWA 600 ppm (1720 mg/m3) OEL-THE NETHERLANDS:TWA 600 ppm (1720 mg/m3) JAN9 OEL-SWITZERLAND:TWA 600 ppm (1720 mg/m3) OEL IN BULGARIA, COLOMBIA, JORDAN, KOREA check ACGIH TLV OEL IN NEW ZEALAND, SINGAPORE, VIETNAM check ACGI TLV

Section 16 - Additional Information

MSDS Creation Date: 3/10/1999

Revision #8 Date: 8/21/2002

The information above is believed to be accurate and represents the best information currently available to us. However, we make no warranty of merchantability or any other warranty, express or implied, with respect to such information, and we assume no liability resulting from its use. Users should make their own investigations to determine the suitability of the information for their particular purposes. In no event shall Fisher be liable for any claims, losses, or damages of any third party or for lost profits or any special, indirect, incidental, consequential or exemplary damages, howsoever arising, even if Fisher has been advised of the possibility of such damages.